Henry Noyes

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Education

Northeastern University — Boston, MA
MS in Robotics | *GPA: 4.00* — *August 2026*
BS in Mathematics and Physics, Minor in Data Science | *GPA: 4.00* — *May 2024*
Honors: NASA Space Technology Graduate Research Fellow, 2022 NASA BIG Idea Challenge Artemis Award, Massachusetts Space Grant Consortium Fellowship, ASME SMRDC 1st Place, Dean's List, National Merit Scholarship
Relevant Coursework: Reinforcement Learning, Control Systems Engineering, Practical Neural Networks, Mobile Robotics, Multimessenger Astrophysics, Machine Learning, Classical Dynamics, Group Theory

Research Experience

Silicon Synapse Lab, Northeastern University — Boston, MA
Student Researcher — *Jan 2022–Aug 2026*

- Co-created the COBRA system (Crater Observing Bio-inspired Rolling Articulator), a modular snake robot capable of sidewinding and tumbling with a mission to measure ice concentration in permanently shadowed lunar craters
- Selected as 1 of 60 national recipients of the 2024 NASA Space Technology Graduate Research Opportunities (NSTGRO) fellowship to advance autonomy in snake robots for lunar exploration
- Served as electrical lead for a team of 11 students selected as 1 of 7 finalists for the 2022 NASA BIG Idea Challenge, where we won the Artemis Award—the top honor of the competition
- Led the lab-wide adoption of modern robotic simulators for reinforcement learning research, creating the open-source tools robot-see and robot-diff to assist in model development
- Developed and deployed a hierarchical control architecture for autonomous whole-body loco-manipulation with a snake robot, demonstrating robust non-prehensile transport of a target object to arbitrary goal locations in a motion capture arena

Robotic Surface Mobility Group, NASA Jet Propulsion Laboratory — Pasadena, CA
Visiting Graduate Researcher — *June–Aug 2026*

- Trained control policies for snake robot locomotion with reinforcement learning in MuJoCo and investigated the impact of contact observations on task performance
- Deployed and evaluated learned policies onboard a snake robot with custom force-sensing modules

Robotics and Intelligent Vehicles Research Lab, Northeastern University — Boston, MA
Robotics Engineer — *Jan–June 2023*

- Led the data processing and system integration efforts on a DARPA-funded mixed reality project focused on perceptually-enabled task guidance (PTG)
- Developed a ROS2 architecture to efficiently collect and store egocentric data from the Microsoft HoloLens 2 headset
- Assisted in training object detection and action recognition models for deployment on an augmented reality headset, and created an automated object annotation pipeline using OptiTrack motion capture

Skills

Languages: Python, C++, MATLAB, Bash, LaTeX
Frameworks/Applications: Isaac Sim/Lab, MuJoCo, ROS/ROS2, Git, Docker, SolidWorks, Blender
Hardware: 3D printing (FDM, SLA), soldering, laser cutting

Publications

A. Salagame, **H. Noyes**, H. Hublikar, and A. Ramezani, "Autonomous Whole-body Loco-manipulation Using a Snake Robot," *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2026, https://arxiv.org/pdf/2611.09777

A. Salagame, **H. Noyes**, E. Sihite, A. Kalantari, and A. Ramezani, "Crater Observing Bioinspired Rolling Articulator (COBRA)," *Advanced Intelligent Systems*, 2025, https://doi.org/10.1002/aisy.202500352